Bihar Engineering University, Patna

B.Tech 6th Semester Exam., 2025

Code: 102605 | ***Subject: Automation in Manufacturing*** | *Time: 03 Hours*

Semester: VI | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) What is the main objective of modeling and simulation in manufacturing?
- (i) To optimize production processes
- (ii) To increase labour cost
- (iii) To eliminate product design
- (iv) To optimize the design

(b) Which aspect of CNC technology must directly enables adaptive control in manufacturing processes?
- (i) Mechanical structure
- (ii) Manual programming
- (iii) Integrated feedback systems
- (iv) Batch processing

(c) In flexible automation, which component is crucial for managing changes in product designs efficiently?
- (i) Rigid fixtures
- (ii) CAD software
- (iii) Assembly lines
- (iv) Hydraulic Systems

(d) What is a distinct advantage of implementing computer graphics software in CAD over traditional design methods?
- (i) Reduced tool wear
- (ii) Lower energy consumption
- (iii) Fast Data transfer
- (iv) Rapid prototyping

(e) Which optimization technique is frequently applied in simulation modeling for manufacturing process improvement?
- (i) Adaptive control logic
- (ii) Process scheduling
- (iii) Linear regression
- (iv) Computer-integrated manufacturing (CIM)

(f) Which of the following is an example of low-cost automation?
- (i) CNC machining center
- (ii) Pneumatic system
- (iii) Computer-aided design
- (iv) Flexible manufacturing system

(g) What is the primary function of PLC in manufacturing automation?
- (i) To design products
- (ii) To control and monitor industrial processes
- (iii) To simulate product routes
- (iv) To handle material manually

(h) What is the main advantage of flexible automation over rigid automation?
- (i) Higher initial cost
- (ii) Requires more human intervention
- (iii) Slower production rate
- (iv) Ability to handle product variety

(i) Which technology is used for automated material handling in flexible automation systems?
- (i) Conveyor belts
- (ii) Robots
- (iii) Pneumatic systems
- (iv) All of these

(j) What does process route modelling help determine in product manufacturing?
- (i) The tool selection for operations
- (ii) The optimal sequence of manufacturing tasks
- (iii) The deployment of worker
- (iv) The machine sequencing

Q.2 Solve both questions : **[7 x 2 = 14]**

(a) What are different types of automation? Explain the need of automation in industries with suitable examples.

(b) Explain low-cost automation.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Discuss the role of CNC and adaptive control in flexible manufacturing systems

(b) Discuss geometric modeling techniques used in CAD for analysis and manufacturing

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Explain the hardware and software components used in Computer Aided Design (CAD)

(b) Explain pneumatic and hydraulic automation with their usage.

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) Explain CNC technology with its application in industries.

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(b) Write short notes on mechanical and electromechanical Systems.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Write short notes on CAD, CAM and CIM.

(b) Discuss the working of PLC and microcontroller with its applications.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Discuss NC part programming with its advantages. Also write any five G-code and its meaning.

(b) Explain process route modelling.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) Discuss the modelling and simulation techniques used in process design

(b) Explain the need and current trends of automation in manufacturing

Q.9 Solve both questions : **[7 x 2 = 14]**

(a) What is an optimization technique? Explain any one with example.

(b) Explain automated material handling system.

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